Improving Chest X-ray Nodule Segmentation via CT-Derived Multi-View Projections

Abstract:
Detecting lung nodules in chest X-ray images remains challenging due to overlapping anatomical structures, low contrast, and limited pixel-level annotations. While computed tomography (CT) offers precise three-dimensional nodule labels, it is less accessible, costly, and involves higher radiation exposure. In this study, we introduce a framework that generates synthetic chest X-ray images and corresponding segmentation masks from annotated CT volumes, enabling effective supervised training for nodule localization. Our approach leverages a U-Net-based architecture augmented with multi-view projections, randomized online augmentations, and a volumetric consistency constraint to enhance robustness, preserve anatomical structure, and improve generalization across varied perspectives. Experimental results demonstrate that models trained on these synthetic radiographs accurately identify lung nodules, with multi-view exposure and advanced augmentations significantly boosting segmentation performance. These findings highlight the value of transferring 3D CT knowledge to 2D X-ray analysis, offering a practical and scalable solution to the scarcity of annotated datasets, and advancing the development of automated, reliable, and accessible lung nodule detection systems.

1. Introduction

1.1 Motivation

Lung cancer is the leading cause of cancer-related deaths worldwide, and survival is strongly dependent on early detection. Large-scale studies, like the National Lung Screening Trial (NLST), show that annual low-dose CT (LDCT) screening reduces lung cancer mortality by roughly 20% in high-risk populations [1], establishing CT as the gold standard. Early nodule identification is critical, as smaller lesions are more treatable and responsive to intervention.

Recent artificial intelligence (AI) advances have further enhanced medical image analysis. Deep learning models trained on CT scans achieve diagnostic performance comparable to experienced radiologists in certain tasks, and some AI-assisted tools have received regulatory approval for clinical use [2]. These developments highlight the potential of automated systems to support early detection and improve clinical decision-making.

Despite CT's effectiveness, the chest X-ray remains a widely used imaging modality in routine practice due to its low cost, reduced radiation exposure, and broad availability. However, X-rays are two-dimensional projections of three-dimensional anatomy. Overlapping structures—such as ribs, blood vessels, and soft tissue—can obscure small nodules, complicating detection and segmentation.

Together, these factors motivate the development of methods to improve lung nodule detection and segmentation in chest X-rays, enabling broader early detection when CT imaging is not feasible.

1.2 Related Work and Research Gap

Deep learning has significantly advanced medical image analysis, including classification, detection, and segmentation. Convolutional neural networks (CNNs) are particularly effective because they can automatically learn hierarchical spatial features. Among these architectures, U-Net has become widely used for biomedical segmentation tasks due to its encoder–decoder structure and skip connections, which preserve spatial detail while integrating contextual information [3].

Many studies have successfully applied deep learning to lung nodule detection and segmentation in CT scans, achieving strong performance on benchmark datasets [4]. The detailed three-dimensional information in CT images facilitates precise annotation and model training. In contrast, fewer studies address segmentation of lung nodules in chest X-rays, largely due to the scarcity of annotated radiograph datasets with pixel-level masks. Producing such annotations is labor-intensive, requiring expert radiologists to outline each nodule, which limits dataset size and diversity.

This contrast highlights a key gap: while richly annotated CT datasets exist, there is no widely adopted approach that systematically leverages these annotations to improve X-ray nodule segmentation. Recently, domain transfer and synthetic data generation techniques—such as Digitally Reconstructed Radiographs (DRRs) and generative adversarial networks (GANs)—have emerged as promising solutions to data scarcity in medical imaging [5, 6]. However, most prior work either focuses solely on CT-based detection or trains on limited real X-ray datasets. The challenge of transferring 3D knowledge to 2D radiographs for precise segmentation remains largely unaddressed. To close this gap, this study leverages CT-derived synthetic radiographs to directly improve segmentation performance and robustness in chest X-ray images.

1.3 Approach Overview

To address the scarcity of annotated chest X-ray data, this study leverages CT volumes with precise 3D nodule annotations to generate synthetic radiographs and corresponding segmentation masks. By projecting nodules from three-dimensional volumes into two-dimensional images, we create labeled training data that enables segmentation learning without labor-intensive manual annotation of real X-rays.

Using this dataset, we train a U-Net-based network to predict pixel-level nodule masks. The training pipeline combines randomized data augmentations with, in certain configurations, a custom volumetric consistency loss that encourages agreement across multiple projections. In addition, we generate synthetic X-ray projections from different viewing angles, exposing the

model to diverse radiographic perspectives. Together, these strategies help the network capture variability in nodule appearance, improve spatial consistency, and learn more robust representations of underlying nodule geometry.

This pipeline allows us to examine whether segmentation models trained on CT-derived synthetic radiographs can effectively localize lung nodules in chest X-ray images, and whether multi-view projections, combined with augmentation and volumetric constraints, enhance segmentation performance and robustness.

1.4 Main Findings

Our experiments demonstrate that CT-derived synthetic radiographs can effectively train segmentation models for chest X-ray images. Multi-view projections and data augmentation improve robustness to variations in nodule appearance, while incorporating volumetric consistency helps maintain structural coherence across views. Overall, these strategies enable the model to localize lung nodules in 2D X-rays more accurately, highlighting the potential of leveraging 3D CT annotations to overcome the scarcity of labeled radiograph datasets.

2. Methods

2.1 Problem Setup and Dataset Construction

We generate synthetic X-ray images from CT volumes using the DiffDRR rendering library, producing paired 2D projection images and segmentation labels derived from the 3D annotations to overcome the scarcity of pixel-wise annotations in chest X-rays. This enables training a segmentation model that can accurately localize lung nodules, capturing their approximate extent and spatial distribution in 2D radiographs.

We use the publicly available NSCLC Radiogenomics dataset, which includes 422 thoracic CT scans with 3D voxel-level nodule annotations. Each CT scan consists of a volumetric intensity image and a corresponding binary mask indicating nodule presence. The dataset is split into 80% training, 10% validation, and 10% test subsets to prevent information leakage.

The following sections describe the generation of multi-view synthetic projections, the U-Net segmentation network, strategies for handling limited data, and an extension for volumetric consistency to improve robustness across multiple views.

2.2 Synthetic X-ray and Label Generation

Given a CT volume and a specified camera pose, DiffDRR simulates X-ray formation by casting rays from a virtual source through the 3D volume toward a detector plane. Each ray traverses the CT volume, and voxel intensities encountered along its path are accumulated to

produce a single pixel intensity value. This process mimics the physical attenuation mechanism of real X-ray imaging.

The projected pixel intensity at detector location (u, v) is expressed as a line integral along the corresponding ray r:

$$I(u, v) = \int_r V(x, y, z)\, ds$$

where V(x, y, z) denotes the CT intensity at spatial location (x, y, z), and ds represents differential distance along the ray. In practice, this integral is computed numerically using the Siddon ray-tracing algorithm, which efficiently calculates intersections between rays and voxel grids.

To generate corresponding 2D segmentation labels, the same ray-tracing procedure is applied to the 3D nodule mask. For each projection ray, if at least one voxel labeled as nodule is intersected, the corresponding 2D pixel is assigned label 1; otherwise, it is assigned 0. Consequently, each CT volume produces a paired synthetic X-ray image and a pixel-wise binary segmentation mask representing projected nodule locations.

2.3 2D Nodule Segmentation Network

We employ the U-Net architecture for pixel-wise segmentation of lung nodules. U-Net is widely adopted in biomedical imaging due to its encoder–decoder structure with skip connections. The encoder progressively downsamples the input through convolutional and pooling layers, capturing increasingly abstract and high-level features. The decoder upsamples these features via transposed convolutions, reconstructing the spatial resolution of the input. Skip connections link corresponding layers in the encoder and decoder, allowing the network to retain fine-grained spatial details that are often lost during downsampling. This enables precise localization of small structures, such as lung nodules, while still leveraging contextual information from larger anatomical regions.

Beyond its segmentation accuracy, U-Net is simple, lightweight, and easy to implement. Its relatively small number of parameters allows effective training even with limited data, and its efficiency enables fast training and inference, which is valuable for iterative experiments with synthetic X-ray images and multi-view augmentation.

Compared to alternative architectures, U-Net provides a strong balance between representational power and computational efficiency. Fully convolutional networks (FCNs) can perform segmentation but often lack multi-scale feature integration, which is crucial for detecting small, subtle nodules in 2D projections. More complex architectures, such as attention-based or transformer models, may offer higher accuracy through long-range dependency modeling, but they require substantially more training data and computational resources. Given the limited number of synthetic X-ray images and variability across projection views, these models risk overfitting, whereas U-Net achieves strong performance with moderate resources.

U-Net's ability to integrate information across multiple scales and maintain spatial precision makes it particularly effective for learning from diverse projection angles. This ensures the model generalizes across variations in nodule appearance, size, and orientation. Its versatility and robustness to data scarcity make it a natural choice for this study, striking an optimal trade-off between segmentation precision, training feasibility, and computational efficiency.

2.4 Overview of Segmentation Pipeline

The full pipeline, from CT volumes to synthetic X-rays with labels, through U-Net to predicted nodule masks, is shown in Figure 1.

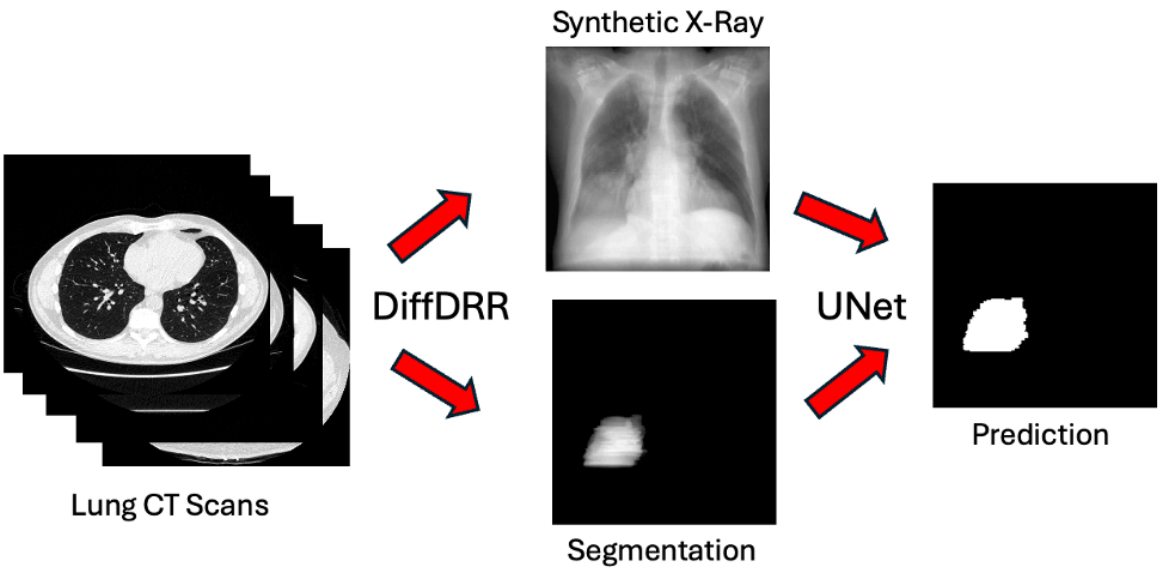

Figure 1: Pipeline Overview

Given an input X-ray image X with height H and width W, the network predicts a probability

map $Y^{2D} \in [0, 1]^{H \times W}$, where each element $Y^{2D}(i, j)$ represents the estimated probability of a nodule being present at pixel (i, j):

$Y^{2D}(i, j) = P(\text{nodule at pixel } (i, j) \mid X)$

Each pixel is treated as a binary classification problem, with a ground truth value of 1 indicating nodule presence and 0 indicating background. During inference, probabilities are thresholded at 0.5 to produce a binary segmentation mask:

$\text{Prediction}(i, j) = 1 \text{ if } Y^{2D}(i, j) > 0.5, 0 \text{ otherwise.}$

2.5 Strategies for Limited Data

To mitigate limited data availability and improve generalization, we employ multi-view expansion together with aggressive randomized online data augmentation.

For each CT volume, two projections (frontal and lateral) are generated, increasing dataset size and introducing viewpoint diversity. In addition, we apply randomized online augmentation using the differentiable computer vision library Kornia. Online augmentation is preferred over offline augmentation because transformations are generated dynamically at every training iteration, eliminating the need to store augmented images while continuously exposing the network to new variations. This stochasticity substantially increases effective data diversity over the course of training. We apply both geometric and photometric augmentation to the images.

Geometric augmentations include horizontal and vertical flips, bounded rotations, and small affine transformations. These transformations primarily alter spatial configuration and are expected to preserve anatomical plausibility; the model should remain robust to such orientation and positioning changes. When geometric transformations are applied to the input image, the corresponding segmentation mask is transformed identically to maintain correct pixel-wise alignment.

Photometric augmentations include mild brightness variation and low-variance Gaussian noise. These are applied more conservatively than geometric transformations because intensity values directly influence pixel-level predictions. Excessive intensity perturbation could distort clinically meaningful patterns; therefore, photometric changes are deliberately constrained to small magnitudes to improve robustness without compromising anatomical fidelity.

In each training iteration, for every original image in the batch, multiple additional independently randomized augmented variants are generated. Thus, each iteration processes the original image plus some newly sampled transformations. For each augmented sample, individual transformations are applied with a probability of 0.5 and if applied, its parameters are also randomly sampled within predefined ranges. For example, rotation angles are randomly sampled within a bounded range and Gaussian noise is sampled from a low-variance distribution.

For even more enhanced augmentation, we additionally generate multiple synthetic projections per CT using diverse camera configurations (different angles and small translation shifts) to further increase exposure to varied viewpoints beyond the standard frontal and lateral projections. Overall, the combination of multi-view generation and heavily randomized online augmentation significantly increases the diversity of training samples and reduces the risk of overfitting despite the limited number of original CT volumes.

2.6 Volumetric Consistency Extension

Projection from 3D to 2D collapses depth information, as multiple voxels along a ray are mapped to a single pixel. To partially reintroduce volumetric reasoning, we incorporate a 3D consistency extension based on geometric backprojection.

Given a predicted 2D segmentation mask, an empty 3D volume with the same spatial dimensions as the original CT is initialized with zeros. For each pixel predicted as nodule (value 1), the corresponding projection ray is traced back into 3D space, and all voxels intersected by that ray are incremented by one.

When multiple views are available, voxels intersected by positive rays from different projections accumulate higher values. Voxels supported by multiple intersecting rays receive stronger geometric evidence of nodule presence. The accumulated volume is then passed through a sigmoid function to obtain a probabilistic 3D prediction. This reconstructed volume can then be incorporated into the training objective as an additional supervision term, encouraging consistency between 2D predictions and anatomically plausible 3D structure.

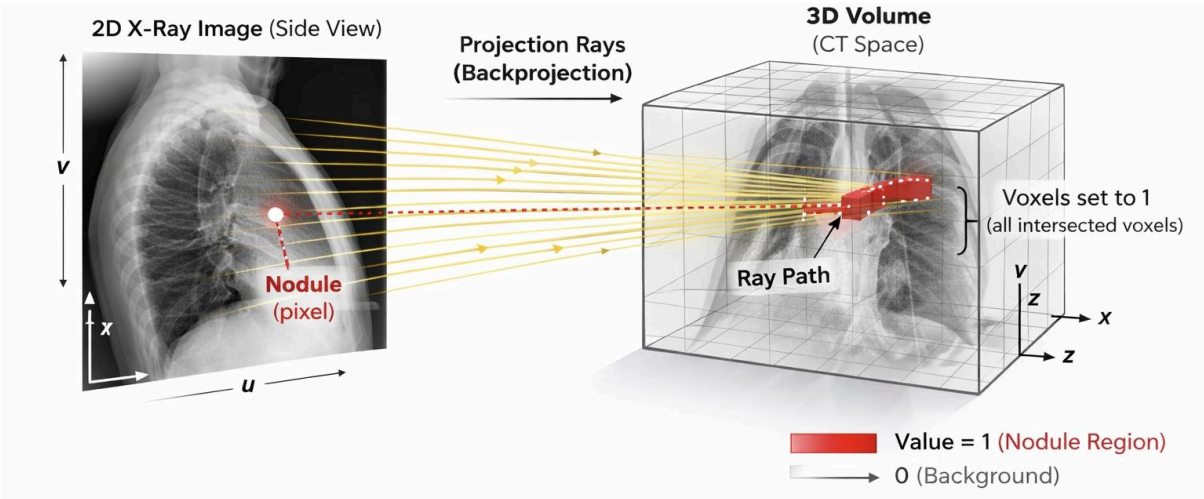

Figure 2: Backprojection of 2D positive pixels into 3D volume

2.7 Training Objective

To train the network, we quantify prediction quality using a loss function. Because nodules occupy a small fraction of pixels, class imbalance is severe. To ensure nodules are detected, we adopt the Dice loss, which emphasizes overlap between predicted and ground truth masks. Unlike standard pixel-wise measures, Dice treats false negatives and false positives asymmetrically, prioritizing correct detection of sparse regions. To prevent excessive false positives and stabilize training, we combine Dice with Binary Cross Entropy (BCE) loss, which measures pixel-wise misclassification.

Let Y_i denote the ground truth label for pixel i, where $Y_i = 1$ indicates the presence of a nodule and $Y_i = 0$ indicates background. Let $\hat{Y}_i$ denote the raw predicted probability at pixel i, produced by the network.

We define a thresholded binary prediction p_i from the probabilities:

$p_i = 1$ if $\hat{Y}_i > 0.5$, else 0

The Dice coefficient measures the overlap between the thresholded predictions pi and the ground truth Yi:

$$\text{Dice} = 2 \times \Sigma_i (p_i \times Y_i) / (\Sigma_i \, pi + \Sigma_i \, Y_i)$$

The corresponding Dice loss is:

$$L_{DICE} = 1 - \text{Dice}$$

Binary Cross Entropy (BCE) is computed directly on the raw predicted probabilities $\hat{Y}_i$ and the ground truth labels Y_i. It is defined as:

$$L_{BCE} = -\frac{1}{N} \Sigma_i \, [\, Y_i \log(\hat{Y}_i) + (1 - Y_i) \log(1 - \hat{Y}_i) \,]$$

where N is the number of pixels.

The combined 2D loss is:

$$L_{2D} = 0.85 \times L_{DICE} + 0.15 \times L_{BCE}$$

Weights reflect the logic of sparse nodule detection: Dice is dominant to prioritize structural overlap, while BCE contributes to pixel-level accuracy and reduces false positives.

Optionally, volumetric consistency can be added via a 3D backprojection. The reconstructed 3D volume V_{PRED} is compared to the original 3D ground truth V_{GT} using BCE:

$$L_{3D} = \text{BCE}(\, V_{PRED}, V_{GT})$$

The final training objective becomes:

$$L_{TOTAL} = 0.8 \times L_{DICE} + 0.1 \times L_{BCE} + 0.1 \times L_{3D}$$

Here, the 3D term acts as geometric regularization, encouraging consistency across multiple X-ray views while keeping the main focus on 2D segmentation. All losses are bounded between 0 and 1, where lower values indicate better agreement with ground truth.

Models are trained for approximately 300 epochs. The final model is selected based on the epoch that minimizes Dice loss on the validation set, ensuring that nodule detection performance is maximized while maintaining generalization. Comparative experiments evaluate models trained with only the 2D loss versus the combined 2D–3D objective to quantify the impact of volumetric consistency under identical data and architectural settings.

3. Results

These experiments evaluate whether 3D CT annotations effectively supervise 2D X-ray segmentation. Our evaluation yields two primary conclusions. First, baseline results confirm

the approach's viability, successfully localizing nodules across varied sizes and projection angles. Second, enhanced multi-view augmentation proves significantly more effective, scalable, and practical for overcoming 2D projection ambiguities than enforcing volumetric consistency. We explore these findings below through quantitative metrics, qualitative visual assessments, component-level analysis, and ROC evaluations.

3.1 Experimental Setup and Evaluation Protocol

Building on the augmentation and training objectives described in Sections 2.4–2.6, we consider three training configurations to study the impact of data diversity and geometric supervision:

- **2D (Moderate Augmentation):** baseline setting using frontal and lateral projections with standard online augmentation (10 randomized transformations per image).

- **2D + 3D Consistency:** same as the baseline, with the addition of volumetric consistency loss.

- **2D (Enhanced Augmentation):** increased augmentation strength (25 transformations per image) and additional synthetic projections from each CT using diverse camera configurations (15 views instead of only frontal and lateral).

Due to the high computational cost of volumetric consistency, we do not combine it with the enhanced augmentation setting.

Performance is evaluated using Dice coefficient and Intersection-over-Union (IoU), reported separately for frontal and lateral projections as well as averaged across both. While the model is trained using the Dice loss, IoU provides a complementary measure of overlap that is widely used in segmentation evaluation and reflects similar trends in performance.

3.2 Quantitative Results

Table 1: Quantitative comparison of segmentation performance (Dice and IoU). Values in parentheses denote standard deviation.

Method	Frontal Dice	Lateral Dice	Mean Dice	Frontal IoU	Lateral IoU	Mean IoU
2D (Moderate Aug)	0.228 (±0.263)	0.254 (±0.279)	0.241	0.157 (±0.197)	0.181 (±0.223)	0.169
2D + 3D Consistency	0.281 (±0.314)	0.254 (±0.301)	0.267	0.209 (±0.253)	0.186 (±0.238)	0.198
2D (Enhanced Aug)	**0.408 (±0.352)**	**0.331 (±0.332)**	**0.370**	**0.321 (±0.294)**	**0.252 (±0.272)**	**0.286**

Overall, we observe substantial variability in Dice scores across all configurations, as reflected in the large standard deviations. This variability is primarily driven by differences in nodule size. Large nodules occupy a significant number of pixels and are generally detected

reliably, yielding high Dice scores, whereas very small nodules often produce near-zero scores when missed. The mixture of these cases leads to high variance in the aggregate metrics.

Across configurations, frontal views generally outperform lateral views, particularly in the best-performing models (2D Enhanced Augmentation and 2D + 3D Consistency). This suggests frontal projections are easier to segment due to clearer structural separation, whereas lateral views introduce ambiguity through greater anatomical overlap. However, this is a general trend rather than a strict rule; occasional anomalies—such as lateral metrics exceeding frontal ones in the 2D Moderate Augmentation setting—likely reflect sample variability.

3.3 Effect of Volumetric Consistency

Incorporating volumetric consistency (2D + 3D Consistency) improves segmentation performance relative to the baseline 2D model with moderate augmentation. The mean Dice increases from 0.2409 to 0.2672, with IoU showing similar increasing trends. However, this improvement is driven almost entirely by gains in the frontal projection, while lateral performance remains nearly unchanged. As a result, the averaged improvement appears relatively modest despite a clear benefit in one view.

As discussed in Section 3.2, frontal projections provide less ambiguous structural information, which likely allows the backprojection-based 3D supervision to more effectively enforce spatial consistency. In contrast, the heavier overlap in lateral views limits the additional benefit from volumetric constraints.

However, this gain comes at a significant computational cost. Training with volumetric consistency is substantially slower (on the order of ~30), which limits scalability and makes it impractical to combine with more aggressive augmentation strategies. In settings where maximizing performance is the primary objective and computational resources are not a constraint, volumetric consistency remains a valuable addition. In practice, however, its high cost makes it less attractive compared to more efficient alternatives.

3.4 Effect of Enhanced Augmentation

Enhanced augmentation substantially improves overall segmentation performance, achieving a mean Dice of 0.3695. Crucially, whereas the addition of volumetric consistency primarily benefited only the frontal projection, enhanced augmentation drives substantial gains across *both* views. Frontal Dice rises to 0.4079, and lateral Dice simultaneously increases to 0.3310, with corresponding improvements in IoU. This demonstrates that increasing data diversity—through aggressive randomization and broader coverage of projection geometries—is highly effective at overcoming the inherent ambiguities of lateral X-rays, yielding a much more balanced and comprehensive performance boost than 3D consistency training alone.

This observation aligns with a broader trend in modern machine learning, where increased data diversity often yields larger gains than added model complexity or more sophisticated training objectives. In this setting, exposing the model to a wider range of projection conditions appears more beneficial than enforcing geometric consistency through reconstruction.

In addition to its performance advantages, enhanced augmentation is computationally efficient. It avoids the substantial overhead of volumetric consistency and scales naturally with available data, making it a practical and effective approach for improving segmentation under realistic resource constraints. It is also significantly easier to implement, as it does not require complex backprojection operations or additional geometric modeling, reducing both development effort and implementation complexity.

3.5 Qualitative Results

While the previous sections focus on aggregate performance metrics, they mask substantial variability across samples, largely driven by nodule size (Section 3.2). To better understand these differences, we examine representative segmentation results for large, medium, and small nodules. In all visualizations below, ground truth is shown in blue, predictions in red, and overlap in purple.

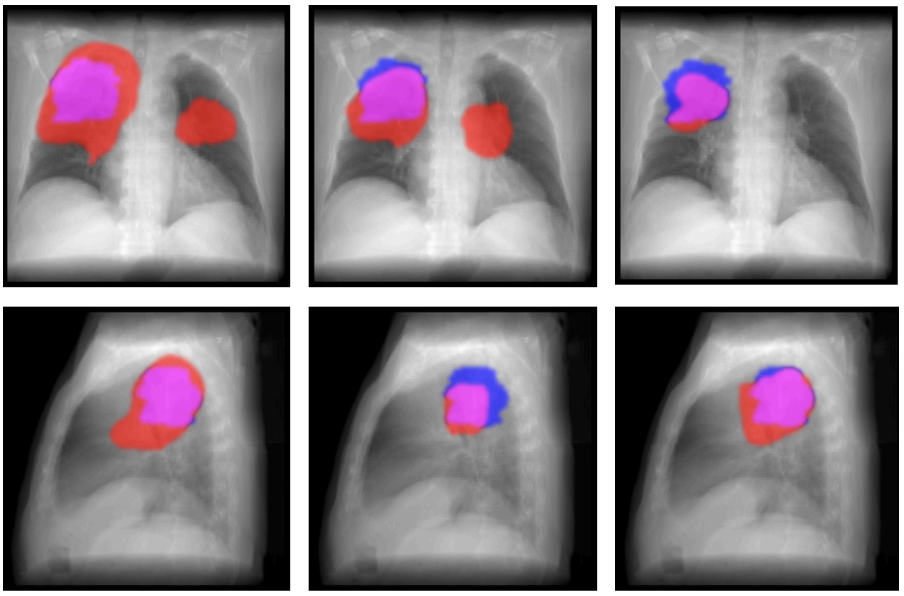

Figure 3: Large nodule segmentation. Top: frontal; bottom: lateral. Left to right: 2D (Moderate Aug), 2D+3D Consistency, 2D (Enhanced Aug). All models localize the nodule, but enhanced augmentation (right) achieves the tightest ground-truth alignment, correcting baseline over-segmentation.

For large nodules, all configurations are generally able to localize the target region, reflecting the relatively strong signal provided by larger structures. However, differences in segmentation quality are still observable across methods. Rather than following a consistent

pattern, these differences vary by case. In the example shown, the enhanced augmentation model aligns more closely with the ground truth, while the baseline overestimates the region. The 2D + 3D consistency model also over-segments, though to a lesser extent.

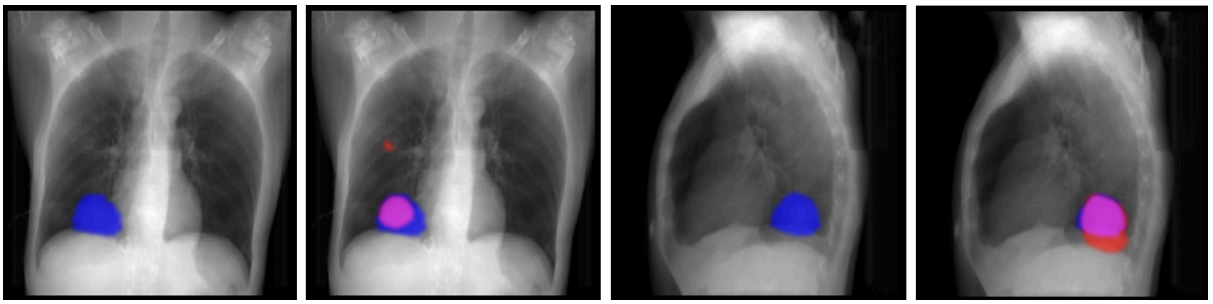

Figure 4: Medium nodule segmentation. Left to right: frontal ground truth, frontal prediction (2D augmented), lateral ground truth, lateral prediction (2D augmented). As nodules shrink, baseline and 3D consistency models often fail; only enhanced multi-view augmentation reliably captures these ambiguous features.

As nodule size decreases, the impact of the training strategy becomes more pronounced. The baseline model frequently fails to detect these nodules, and while 3D consistency can provide modest improvement in some cases, in this example only the 2D (Enhanced Augmentation) model successfully identifies the nodule, with both the baseline and 2D + 3D consistency models producing no detection. This highlights the importance of diverse projection exposure in handling more ambiguous cases.

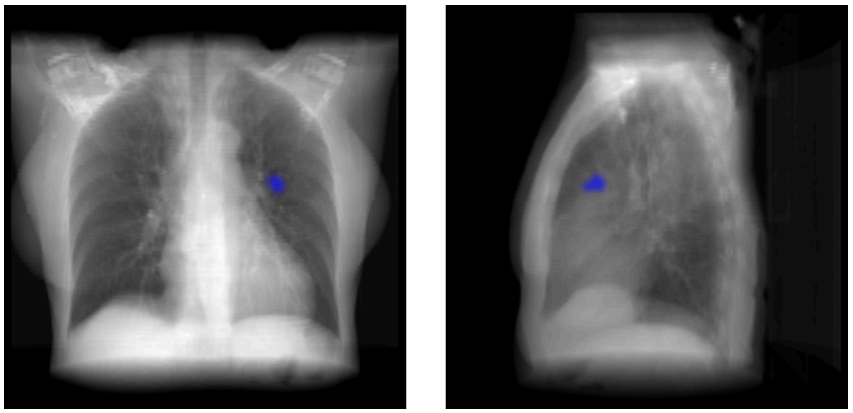

Figure 5: Very small nodule failure case. Frontal (left) and lateral (right) ground truths; no models produced predictions. Losing depth information in 2D projections consistently causes all configurations to miss highly obscured, low-pixel nodules.

For very small nodules, all configurations struggle, frequently producing false negatives. This is expected given the limited resolution of 2D projections and the loss of depth information during projection. In such cases, nodules occupy only a few pixels and may be heavily obscured by overlapping structures, making reliable detection inherently difficult.

3.6 Component-Level Detection Analysis

To better evaluate practical detection performance, we perform a component-level analysis, where each connected region in the predicted and ground truth masks is treated as an individual object. This shifts evaluation from pixel-wise agreement to object-level detection, focusing on whether the model correctly identifies distinct nodules rather than only matching their spatial overlap. Unlike pixel-wise metrics such as Dice or IoU, which measure similarity between continuous regions, this formulation directly evaluates detection of individual nodules.

We define true and false positives using a bidirectional matching strategy. From the prediction perspective, each predicted connected component is counted as a true positive if it overlaps with any ground truth nodule by at least 50% of its area; otherwise, it is a false positive. From the ground truth perspective, each annotated nodule is considered detected if at least one predicted component overlaps with at least 50% of its area; otherwise, it is a false negative. This dual counting scheme ensures that both over-detection (false alarms or fragmented predictions) and missed detections are properly captured. Although this procedure defines true positives from two perspectives, they are used jointly to compute a single set of evaluation metrics—precision, recall, and F1-score. Specifically, prediction-based counts determine precision while ground-truth-based counts determine recall. The resulting metrics summarize detection performance at the object level in a clinically meaningful way.

Table 2: Component-level detection performance

Model	Precision	Recall	F1 Score
2D (Moderate Aug)	0.251	0.411	0.309
2D (Enhanced Aug)	**0.483**	**0.420**	**0.448**
2D + 3D Consistency	0.463	0.305	0.365

As shown in Table 2, enhanced augmentation achieves the best overall detection performance, attaining the highest F1 score while maintaining a strong balance between precision and recall. In contrast, the baseline model exhibits higher recall but substantially lower precision, indicating a tendency to produce excessive false positives. The 3D consistency model achieves relatively high precision but lower recall, reflecting more conservative predictions that miss a larger fraction of nodules.

The asymmetry between prediction-centric and ground-truth-centric true positives provides additional insight into model behavior. In the 3D consistency model, the number of prediction-based true positives exceeds ground-truth detections, suggesting that some predicted regions overlap only partially with nodules, often producing coarse or overly large segments. In contrast, enhanced augmentation shows a closer alignment between these measures, indicating more accurate and well-localized predictions.

3.7 ROC Analysis

While the component-level analysis evaluates discrete nodule detection, it depends on a fixed binarization threshold. To complement this, we analyze model performance across all possible thresholds using Receiver Operating Characteristic (ROC) curves. This provides a threshold-independent view of the trade-off between sensitivity and false positive rate, offering a more complete characterization of detection behavior.

Specifically, we compute ROC curves separately for frontal and lateral projections, alongside a combined setting. True negative samples are obtained from an LIDC-IDRI subset with a multi-radiologist consensus of zero nodules. While clinical datasets rarely guarantee absolute purity, this highly confident subset is similarly projected into DRR X-rays to enable explicit evaluation of false positive behavior. The area under the ROC curve (AUC) summarizes overall discriminative ability, while the optimal operating point is selected based on the threshold maximizing validation performance.

The results show that the model achieves stronger discrimination in frontal views compared to lateral views. The frontal projection yields an AUC of **0.810**, indicating relatively reliable separation between nodules and background structures. In contrast, the lateral view achieves a lower AUC of **0.705**, reflecting the increased ambiguity and anatomical overlap inherent in this projection. When combining both views, the AUC is **0.748**, suggesting that while multi-view aggregation provides additional information, its benefit is limited by the weaker performance of the lateral projection.

An important observation is that the optimal decision thresholds are very high (approximately **0.96–0.99** across settings). This indicates that the model tends to produce overconfident predictions, requiring aggressive thresholding to suppress false positives. This behavior is consistent with the component-level results, where models with higher recall often exhibited lower precision due to spurious detections.

Overall, the ROC analysis reinforces the findings from the object-level evaluation. Frontal views remain the most informative for detection, while lateral views introduce additional challenges that are not fully mitigated by simple combination. Moreover, the need for high thresholds highlights a calibration issue, suggesting that future improvements could focus on better probability calibration or more robust decision rules.

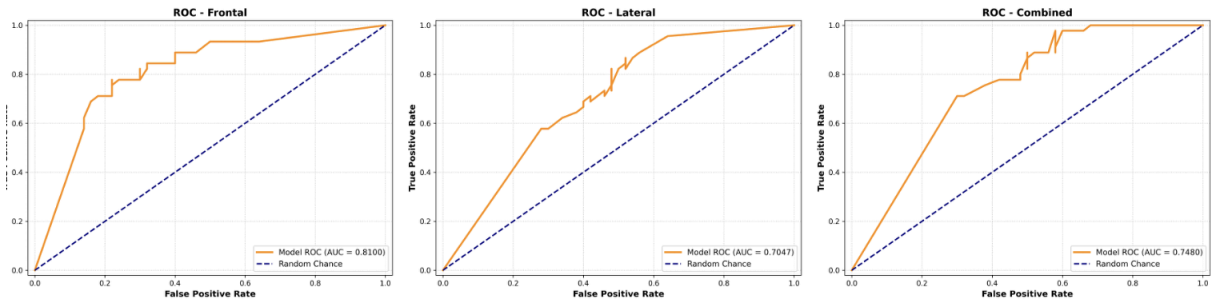

Figure 6: ROC curves across projections. From left to right: frontal, lateral, and combined views. Frontal views exhibit meaningfully stronger performance than lateral perspectives.

3.8 Discussion and Practical Implications

The results highlight a trade-off between geometric modeling and data-driven augmentation. Volumetric consistency improves performance, particularly in frontal views, by enforcing 3D structural constraints, but its impact is uneven across projections and comes with substantial computational cost. Although it explicitly reconstructs a 3D volume, it relies on a limited number of views, constraining the geometric information it can recover. In contrast, enhanced augmentation achieves stronger overall performance by increasing data diversity through both transformations and varied projection geometries. This suggests that, in this setting, directly exposing the model to a wider range of projection conditions is more effective than imposing consistency through reconstruction, likely because it better captures the variability inherent in X-ray imaging. As noted in the introduction, the gold standard remains full 3D information; increasing the number and diversity of 2D projections brings the model closer to this, effectively exposing more of the volumetric structure and enabling implicit reconstruction via multi-view supervision.

These differences are particularly relevant when considering how the model is used in practice, where efficiency and robustness often matter more than perfect reconstruction fidelity. From an application perspective, the objective is not perfect pixel-wise segmentation but reliable localization. Even coarse predictions can highlight suspicious regions for further review. While very small nodules remain challenging, the model performs effectively on larger and moderately sized cases, where approximate localization is often sufficient to support downstream analysis.

4. Conclusion

This study addresses the challenge of accurately detecting lung nodules in chest X-ray images, where pixel-level annotations are scarce and 2D projections inherently lose critical volumetric information. We leverage annotated 3D CT volumes to generate synthetic radiographs with corresponding segmentation masks, enabling high-quality supervised training for 2D nodule segmentation. To mitigate overfitting and compensate for information loss in 2D projections, we incorporated randomized data augmentation, multi-view projections, and a volumetric consistency constraint. Crucially, our findings demonstrate that enhanced multi-view augmentation is a significantly more effective and practical solution than enforcing geometric constraints via 3D backprojection. By exposing the model to a wider array of projection geometries, it successfully learns robust spatial representations from limited data without the prohibitive computational overhead of volumetric reconstruction.

Despite the reduced dimensionality of X-ray imaging, this approach enables fast, low-cost, and widely accessible nodule detection while maintaining reliable localization of clinically relevant nodules. To build on these findings, future work must focus on bridging the sim-to-real domain gap by evaluating this framework on diverse, real-world clinical radiograph datasets. Additionally, while multi-view augmentation proved most efficient, future iterations could explore integrating computationally lightweight sparse backprojection

techniques or self-supervised contrastive learning to harness 3D structural priors more scalably. Overall, our framework demonstrates that 3D CT knowledge can be effectively leveraged to enhance 2D X-ray segmentation, providing a practical, scalable, and robust solution for automated lung nodule detection.

References

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